Reading the Computation: How Compression Regime Determines Whether Lost Memory Is Routable

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Artifacts: `memsub-artifacts:/ {gate-eval, compactor-certification, probe}`

Abstract

Lossy compression of a language model's context memory produces silent failures: facts the model once held are answered wrongly, with full confidence, and no signal at generation time. We show that *which* signals can predict — and therefore route around — this loss depends on the compression regime, and we measure the full picture on one certification protocol, one frozen 4B model, and a ∼1B-token four-domain corpus of 98,355 counterfactually filtered probes. Under **contiguous eviction**, loss is governed by observable mechanics: a ≈$140 dual-encoder gate over text shadows cuts silent loss 72% at a fifth of always-expand cost, ECE ≤ 0.023. Under a **trained compactor**, the same gate class collapses to surface-statistics performance (AUC 0.55 ≈ a five-feature logistic at 0.56): the loss becomes invisible to write-side and corpus-side signals. It does not become unpredictable — it becomes *internal and query-conditioned*. A linear probe on the frozen model's residual stream (layer 30), with the question present, predicts answer divergence at 0.819; a small stack reaches 0.854 and, deployed as a gate, matches the shadow gate's correctness at one third of its expansion budget. A context-only control shows the signal is *relational*: without the question, the same layer carries no linearly decodable trace (AUC 0.502) — consistent with sufficiency being a property of (query, memory) pairs rather than of the memory state alone. We further show (i) a deployment ladder of increasing cost — a prompted self-check (0.638, one forward, no training), a two-hook attention-head monitor (0.762), and the trained probe stack (0.854); (ii) that probe scores rank an instance-independent *fragility ordering* validated across compactor reseeds (κ=0.745; Kendall τ 0.57); and (iii) a mechanistic surprise: our 1/40-scale compactor's cross-attention consists of exact positional delta kernels — it performs strided subsampling, not content synthesis (**position-kernel collapse**; the learned kernel is a uniform stride-3 lattice, identical across heads and items, and its geometry is loss-blind — needle-to-kept-position distance predicts survival at 0.509), which reframes both its certification failure and its unexpectedly *better* pointwise retention versus eviction at matched budget (0.27 vs 0.58 silent loss). Total marginal compute across all experiments: ≈$1,100.

1 Introduction

"Long context" conflates a serving problem—fitting the tokens—with two capability problems that separate the moment full attention becomes unaffordable: *pointwise* recall of exact, low-redundancy facts, and *aggregative* understanding of distributed, redundant content. A needle-class query—an identifier, a figure, a verbatim clause—concentrates its answer in exact bytes with near-zero redundancy; any lossy transform destroys it at a rate proportional to the compression ratio. An integration-class query lives on distributed, redundant information; it is compression-tolerant and often mildly compression-*benefited*. The two are not merely different but antagonistic: compression bought for the aggregative workload is paid for in pointwise fidelity.

The antagonism makes a *router* the critical component: a calibrated per-query estimate of whether compressed memory suffices. This paper's question is what information such a router can be built from, and the answer turns out to be regime-dependent in a way we believe has not been measured before. Delete tokens, and the damage follows mechanics anyone can see from bookkeeping features; subsample-and-smear, and the damage moves inside the model, and further in than that—it is not stored in the memory state at all, but materializes in the encounter between query and memory, where a linear probe on the residual stream recovers what every external signal missed.

Contributions: **(i)** a two-axis certification protocol (aggregation / pointwise, never blended) with counterfactually filtered probes; **(ii)** its application to two compression regimes at matched retained budget, showing the axes *trade*—eviction preserves aggregation and destroys needles; our compactor degrades aggregation while halving silent needle loss; **(iii)** the routability result: eviction loss is mechanics-routable (0.72 reduction from shadow features), compactor loss is routable only by reading the computation (0.85 AUC from the residual stream; 0.56 from everything outside it); **(iv)** a three-rung deployment ladder with measured cost/benefit at each rung; **(v)** position-kernel collapse as a failure mode of small-scale learned compaction, with cross-attention entropy as its certification-time diagnostic; **(vi)** instance-independent fragility ordering; **(vii)** multi-hop entry-gating beating per-block selection (1.98× vs 2.33× oracle cost).

2 Related work

KV-cache reduction. Per-context selection scores tokens at inference and evicts the rest (H2O [14], SnapKV [8], StreamingLLM [12]); amortized selection learns the scorer (KV-Distill [3]); per-context and amortized synthesis write new compact entries rather than keeping original rows (Still [9]); architecture-level lines bake reduction into the model (MLA [4], native sparse attention [13]). Our work is orthogonal to the choice of compressor: we certify compressors along two axes and supply the decision layer their lossiness requires. The strongest published synthesis results themselves show the axis split we formalize—near-full aggregative utility with needle-retrieval collapse—which we take as independent evidence for the two-workload framing.

Selective prediction and abstention. The gate is a selective-prediction head [5] over an external memory: it estimates, per query, whether the model's conditioning state supports the answer. The label is *counterfactual* (answers under compressed vs. raw conditioning are compared, rather than correctness being judged in isolation), and the head is trained fully decoupled from the backbone. Calibration machinery follows Guo et al. [6] and Platt [11].

Probing and representation reading. Linear probes on intermediate activations read what a network has computed [1]; recent work reads truthfulness and uncertainty from residual streams [2]. Our probe is a *sufficiency* probe over an external memory tier, supervised by counterfactual behavioral labels rather than truth labels, and—as our context-only control shows (§6.4)—it reads a query-conditioned divergence signal, not a static property of the stored representation.

Attention sinks and memory-augmented agents. We repurpose attention-sink mass [12]: under-attention to memory parks probability on sink positions, so sink-mass anomaly becomes an insufficiency sensor (§6.3). OS-style memory hierarchies (MemGPT-class [10]) page text in

and out of context but do not certify fidelity or route by workload; blended long-context suites (RULER [7]) are exactly the evaluations our protocol unbundles.

3 Two axes and a certification protocol

Definitions. Fix a base model f, a compression policy C, and a corpus of (context, question, answer) items where each question is generated from a specific source span. An item is *counterfactually valid* if f answers it correctly with the context and incorrectly without it—so a correct answer under compression proves the information traversed the compressed cache. The **aggregation axis** is measured over integrative behavior: next-token KL between f conditioned on the full cache and f conditioned on C(cache), plus answer-accuracy deltas on items whose source spans survive compression. The **pointwise axis** is the *lost-fact incident rate*: among counterfactually valid items whose source span was compressed away, the fraction answered incorrectly *without* any expansion mechanism—the silent-failure rate.

Protocol. A configuration is certified per axis, never as a blend; the noise floor is established by running the identical probe battery at zero compression (measured 0.000); and a configuration certified only on the aggregation axis may ship only behind an active, calibrated expansion gate for pointwise traffic. Every report object embeds an evaluation-protocol record: the exact item partition per axis, the contents of the compacted context during probing (surviving KV only; no summary text), confirmation that expansion was disabled, and the scoring rule (argmax over answer-letter logits, greedy—format failures structurally impossible).

4 Regime I — contiguous eviction: mechanics-routable loss

4.1 Setup and axis split

Base model Qwen3-4B-Instruct-2507 (bf16, greedy, pinned runtime fingerprint). Corpus: 40,000 long documents across financial filings (238M tokens), literature (416M), case law (119M), and source code (154M); 98,355 extractive multiple-choice items generated by the base model from random ∼1k-token sub-chunks, teacher-verified answerable with context and discarded if answerable without. Compression policy: evict-oldest-non-sink at fractions 0.25 and 0.40, first four positions pinned as sinks [12]; per-item 4096-token prefills with hash-randomized placement.

The zero-eviction control gives a lost-fact noise floor of exactly 0.000 (conditioned on prior-correct). Under eviction the aggregation axis barely moves—mean KL 0.0303 (f=0.25) and 0.0339 (f=0.40)—while the pointwise axis collapses: **0.453** and **0.551** of facts in evicted spans are silently lost. A matched-random control (same token count evicted uniformly) inverts the damage profile—4× less pointwise loss at 3× the aggregation KL—so recency-structured policies concentrate pointwise damage exactly where aggregate metrics cannot see it. One artifact, two workloads, opposite outcomes.

4.2 The shadow gate

For each item and eviction fraction we produce a counterfactual sufficiency label (y=1 iff the compacted-cache answer differs from the raw-span answer; 196,710 rows). The gate is a dual encoder over frozen Qwen3-Embedding-0.6B with a 256-d normalized projection interaction, a small feature head (eviction fraction, query–block overlap, evicted flag, domain), and an

Table 1: Signal decomposition (3 seeds, mean ± SD). Features-only matches the full model on the evicted-only task; embeddings add essentially nothing.

Variant	evicted-only AUC	evicted-only PR-AUC	pooled AUC
full model	0.624 ± 0.004	0.622	0.831
features-only (encoders zeroed)	0.621 ± 0.006	0.620	0.837
embeddings-only (features zeroed)	0.536 ± 0.004	0.561	0.557

Table 2: Matched-expansion-rate comparison (test split, lost-fact incident rate conditioned on prior-correct).

Expansion rate	0.1	0.2	0.3	0.4	0.5
gate	0.277	0.092	0.000	0.000	0.000
query-type routing	0.470	0.416	0.368	0.308	0.248

auxiliary query-workload head; a frozen-embedder linear probe sufficed. It is natively calibrated (per-fraction T=1.0; test ECE 0.023/0.017); marginal cost ≈$140.

Pooled test AUC is 0.829, but pooling conflates separating evicted from survived rows (near-deterministic) from the question that matters—among evicted spans, which lost their fact. Restricting to evicted-span rows, the full model reaches AUC **0.622** ($[0.608, 0.636]$), but the features-only ablation reaches 0.621 ± 0.006—within seed noise of the full model (0.624 ± 0.004)—while embeddings-only falls to 0.536 ± 0.004 (Table 1). **The signal is eviction mechanics**, and that is enough here. At τ=0.488 the gate reaches test lost-fact incident rate 0.146 ($[0.136, 0.156]$) at expansion 0.168 ($[0.163, 0.173]$), against never-expand 0.453–0.551 and oracle 0 at 0.194, strictly dominating query-type routing at every matched rate (Table 2, Figure 1).

4.3 What travels and what doesn't

Under leave-one-domain-out evaluation the gate's discrimination holds (pooled AUC 0.781–0.865) but calibration degrades to ECE 0.05–0.13. A scalar temperature alone left three of four domains above the 0.05 bar; escalating to Platt scaling (temperature + bias on frozen scores) restores ECE to 0.014–0.035 in every rotation (Table 3). A "calibrated gate" is a checkpoint plus a per-domain recalibration procedure—retained as the deployment procedure for every gate tier in this paper (Figure 2).

5 Regime II — a trained compactor: the axes cross, the gate goes blind

5.1 Compactor, certification, no-ship

Still-canonical per-layer Perceiver (per-KV-head latent banks, d_l=256, two pre-norm QK-normed cross-attention blocks, no FFN, ≈50M parameters) trained at 1/40 the published data scale against the frozen base, certified on the same protocol at matched retained budget (sinks + t=1024 latents + 1,024 recent raw slots of a 4,096-token context). Aggregation axis: *worse* than eviction (KL 0.263 vs 0.122; recurrence depth 4 collapses to 2.29 → `max_certified_depth` = 0).

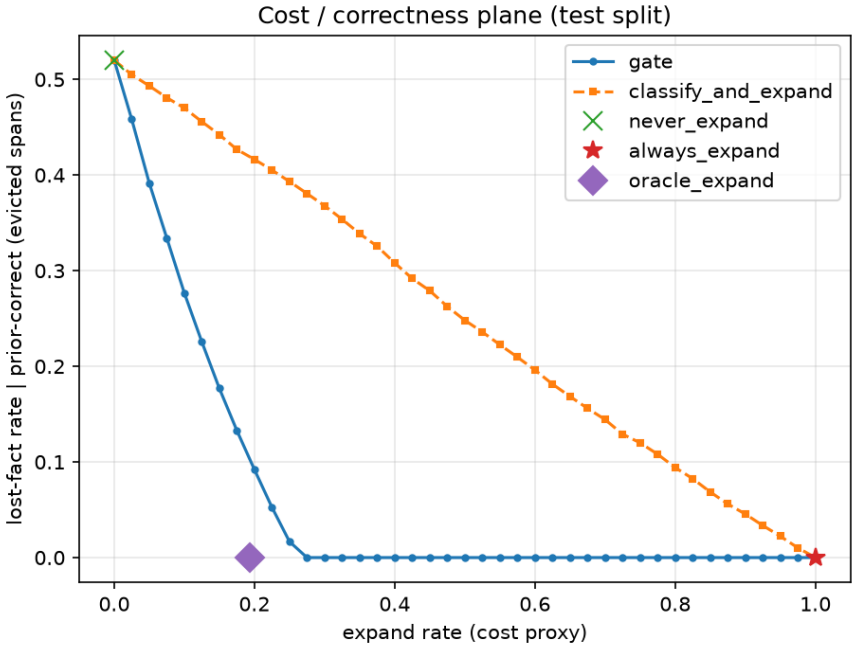


Figure 1: Cost–correctness plane (eviction, test split). The gate curve dominates query-type routing everywhere; oracle at $(0.194, 0)$; never-expand at $(0, 0.52)$ pooled.

Table 3: Leave-one-domain-out recalibration on ∼200 items from the held-out domain (frozen gate scores).

Held-out domain	method	ECE before	ECE after
financial	Platt	0.130	0.014
legal	Platt	0.106	0.029
code	temperature	0.064	0.035
literature	Platt	0.116	0.023

Pointwise axis: *better*— silent needle loss 0.271 vs eviction's 0.583 (Table 4). The regimes trade axes at this scale; both numbers sit far above the 0.000 floor, so the gate-mandatory rule re-attaches under every compaction regime measured.

5.2 Position-kernel collapse

Captured cross-attention is exactly positional: delta rows, entropy 0.000 across 1,000 items × 36 layers × 8 heads. The compactor never left its position-dominated initialization—compaction here is strided subsampling of $[K; V]$ plus leakage smear, not content synthesis. Reading the per-latent argmax source positions directly, the kept set is exactly round$(k \cdot (S-1)/1023)$, k=0..1023: a uniform stride-≈2.998 lattice at source length S=3068 (density 1/3), *identical across all 8 KV-heads and all items* (item-to-item difference 0). This (a) renames the regime honestly (subsampling, with true synthesis untested at this scale); (b) explains the pointwise advantage (subsampled latents smuggle partial content forward; deletion destroys it); (c) explains the cross-reseed nesting of failure sets (§7); (d) yields a certification diagnostic: **report compactor cross-**

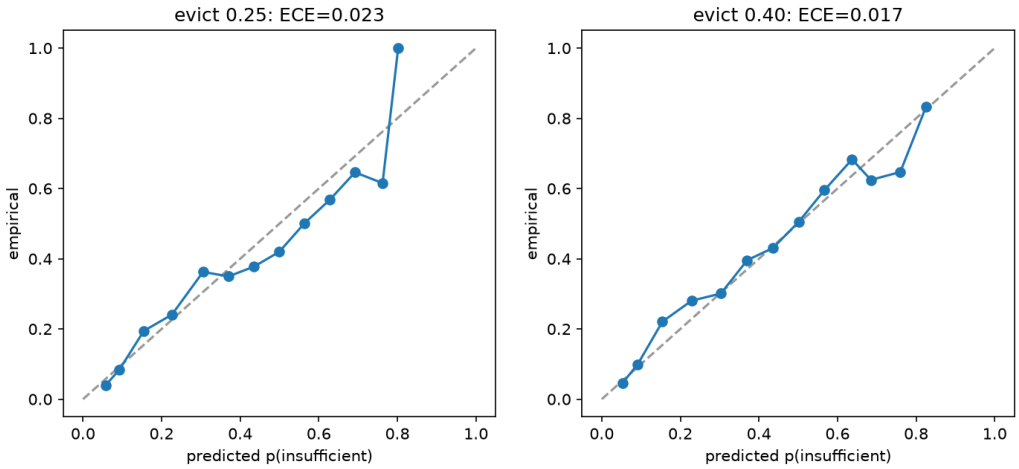


Figure 2: Reliability curves per eviction fraction; the linear probe is natively calibrated (T=1.0; ECE 0.023/0.017).

Table 4: Trained compactor vs. untrained eviction at matched retained budget (12 contexts across 4 domains). All rows on the CERTIFICATION protocol (needle-anchored contexts); do not compare against labeling-protocol numbers — see Appendix B.

arm	mean KL	acc. delta	lost-fact | prior-correct
control (no compression)	0.000	0.000	0.000
eviction @ matched budget	**0.122**	−0.049	0.583
compactor, depth 1	0.263	−0.090	**0.271**
compactor, depth 4 (recurrence)	2.292	−0.056	0.750

attention entropy; a near-zero value means the artifact is a subsampler regardless of its loss curve (ours hit KL 0.058 by step 300 while never routing content). We name the failure mode *position-kernel collapse*, the compactor-side sibling of recurrent-gate collapse.

5.3 The gate goes blind

A new compaction policy is a new label distribution, so we relabel all 98,355 items under the compactor (positive rate drops to 0.167) and retrain the shadow gate. The signal decomposition of §4.2 *flips*—embeddings-only (0.567) now beats features-only (0.537)—but everything lands at ≈0.55, statistically indistinguishable from a five-feature surface logistic (0.564 $[0.553, 0.575]$); engineered interference-geometry features add nothing (0.568). The refreshed gate still *calibrates* (ECE 0.0074) and rides the lower base rate to a nominal operating point, but its marginal reduction is 22.5% at 86% of oracle's budget: it is calibrated always-expand-lite, not a router. **The loss did not become smaller-and-harder; it became internal** (Table 5).

Table 5: The routability asymmetry (labeling protocol, test split). "Reduction" is the gate's marginal effect vs. never-expand under the SAME policy; "budget use" is gate expansion over oracle expansion.

regime	never	gate @ τ	oracle	reduction	budget use	evicted-only AUC
eviction	0.520	0.146 @ 0.168	0 @ 0.194	**72**%	0.87	0.62 (mechanics)
subsampling	0.189	0.146 @ 0.147	0 @ 0.170	**22.5**%	0.86	0.55 (content)

Table 6: Sufficiency signals under the subsampling regime (evicted-only, doc-level test split; τ = Kendall rank correlation with the cross-reseed fragility ordering).

signal source	AUC [95% CI]	τ
surface features (floor)	0.564 [0.553, 0.575]	—
+ interference geometry	0.568 [0.557, 0.579]	—
prompted self-check (1 fwd)	0.638 [0.628, 0.649]	0.20
sink/retrieval-head monitor (2 hooks, 8 scalars)	0.762 [0.753, 0.771]	—
per-head attention probe	0.767 [0.758, 0.777]	0.46
residual linear, layer 30	0.819 [0.811, 0.827]	0.45
residual MLP (3 seeds)	0.852 [0.844, 0.859]	0.47
stack	**0.854** [0.846, 0.862]	**0.57**

6 Reading the computation

6.1 Probes

All probes trained on the same counterfactual labels, evicted-only evaluation, doc-level splits, fixed capacity, cluster-bootstrap CIs; the fragility-ordering row (Kendall τ vs the cross-reseed nesting, §7) is reported for every arm.

6.2 Results

Table 6: the layer curve is monotone through the last third of the stack (0.53 at L15 $\rightarrow$ 0.65 $\rightarrow$ 0.81 $\rightarrow$ peak L30)—sufficiency estimation crystallizes late. Aggregated attention statistics fail (best 0.536) while individual late-layer heads reach 0.72—including a cluster whose signal is *anomalous sink mass*: under-attention to memory parks probability on sink positions, turning the attention-sink phenomenon into a sensor.

6.3 The deployment ladder

Three rungs, each with its measured cost: the prompted self-check (0.638; one extra forward, zero training—already above every corpus-side approach); the head monitor (0.762; end-to-end lost-fact incident rate 0.149 at expansion rate 0.060; two scalar hooks on one forward, no capture infrastructure—and notably the prompted self-check adds nothing on top of it (0.763 combined): the model's verbalized self-report is subsumed by its own attention behavior); the trained stack (0.854; capture + $3 of probe training per capacity point). End-to-end at natural prevalence, the stack-gated system reaches lost-fact incident rate 0.146 at expansion rate **0.050**—the shadow gate's correctness at one third of its budget, strictly inside its curve, with oracle at 0/0.170.

Per-capacity retraining is required (transfer to t=512 drops 0.21, consistent with cross-capacity label κ 0.455); per-domain temperature repair carries over from §4.3.

6.4 What the probe reads

A context-only control—the same compact context, no question, same layer—drops to chance: a fresh probe reaches 0.502 $[0.487, 0.518]$ and the question-conditioned probe applied zero-shot reaches 0.508 $[0.493, 0.523]$, against 0.813 with the question present. The probe is therefore a **behavioral-divergence predictor**—it reads the model's answer-time state, question included—and should not be interpreted as a readout of memory contents. We regard this as confirmation rather than deflation: our certification protocol *defines* sufficiency counterfactually, as a relation over (query, memory) pairs, and the control shows the model's internals agree—there is no query-independent sufficiency signal to read, because sufficiency is not a property the memory state has on its own. (Scope: the null is for linear decodability at the probed layer; nonlinear or multi-layer context-only readouts are untested.)

7 Fragility is a property of the fact, not the checkpoint

Cross-reseed relabeling: κ 0.745 with the weaker instance's failure set nested inside the stronger's (P=0.945) and protocol determinism exactly 1.0—loss follows an instance-independent fragility ordering that better compactors peel from the top. Probe scores rank this ordering (τ 0.57, CI excludes 0). Qualitatively, fragility concentrates in key→value *binding* under confusable-sibling pressure— neighboring table rows, near-identical identifiers, common-token values in crowded neighborhoods—with token rarity mildly *inversely* predictive: what dies is binding, not content.

8 Discussion

8.1 The routability ladder

One protocol, two regimes, three signal classes:

eviction	loss follows visible mechanics → shadow gate: −72% loss
subsampling	kernel geometry loss-blind → shadows blind (0.55),
(this work's "compactor")	kernel mechanics blind (0.55); query-conditioned computation-reading 0.85 → same loss at $\frac{1}{3}$ budget
true synthesis	untested at this scale

8.2 Is subsampling loss mechanics-routable after all?

No—and the test was as favorable as it could be made. The kernel is fully known (the exact stride-3 lattice, item-invariant), yet needle-to-nearest-kept-position distance predicts survival at 0.509 $[0.499, 0.519]$, the mechanics-with-kernel gate adds +0.003 over the surface floor (0.550 $[0.537, 0.563]$), and the residual-stream probe's scores correlate with kernel distance at ρ=0.045—almost none of the 0.85 is kernel geometry. Because every span of three or more tokens contains a lattice point, what varies is only the phase of the answer tokens relative to the lattice, and phase carries no signal: survival is decided by leakage smear and by how downstream layers use the subsampled cache. The internality claim stands in full, with one refinement from the context-only control: the loss is not merely internal to the model—it is internal to the *query-conditioned*

computation. Write-side blind, question-free read-side blind; only the encounter between the question and the memory carries the signal.

8.3 Consequences for memory-system design

Gate-mandatory is regime-independent; gate-*class* is regime-dependent. Certification must include the probe: any retrained compactor gets its divergence-predictability measured per instance and per capacity, its cross-attention entropy reported, and—if the kernel is nontrivial—the kernel-geometry test rerun (a full-recipe compactor that escapes position-kernel collapse may make write-time geometry loss-bearing again; ours did not). Multi-hop: content-aware *entry* (per-query sufficiency score) beats per-block ranking (1.98× vs 2.15–2.33× oracle)—the binding constraint was never block scoring.

9 Limitations

Single 4B base; one tokenizer (fragility neighborhoods are partly tokenizer artifacts); the compactor is 1/40-recipe and mechanistically a subsampler—Regime II results certify this artifact class, not full-recipe synthesis. A post-hoc review also found the shipped checkpoint's training loss used a support-construction variant that double-counted the gold token whenever it was already in the teacher top-200 (corrected in the released code); all certified numbers measure the checkpoint as trained. Extractive MCQ probes, letter-match equivalence, greedy decoding, single-turn contexts; multi-hop k=2, one evicted candidate per query; the end-to-end routability point is weighted at natural prevalence on the captured sample; F1's needle features exclude the 22% of test items without a locatable in-prefix answer; the F2 null is scoped to linear decodability at layer 30. Probe claims are per-capacity and per-instance by measurement (§6.3, §7), and the probe predicts behavioral divergence rather than certifying memory contents (§6.4).

10 Conclusion

Silent memory loss under compression is not one problem. Delete, and the damage follows mechanics anyone can see; subsample-and-smear, and the damage moves inside the model—and further in than that: it is not stored in the memory state at all, but materializes in the encounter between query and memory, where a linear probe on the residual stream recovers what every external signal missed, for a third of the routing budget. The practical statement is a ladder—ask the model (free), watch two attention heads (cheap), read the residual stream (small)—and the scientific one is that *routability, not just fidelity, is a certifiable property of a compression regime.* We think that is the property long-context memory systems should be certifying next.

References

[1] Guillaume Alain and Yoshua Bengio. Understanding Intermediate Layers Using Linear Classifier Probes. In *International Conference on Learning Representations (ICLR) Workshop*, 2017.

[2] Amos Azaria and Tom Mitchell. The Internal State of an LLM Knows When It's Lying. In *Findings of the Association for Computational Linguistics (EMNLP Findings)*, 2023.

[3] Vivek Chari, Guanghui Qin, and Benjamin Van Durme. KV-Distill: Nearly Lossless Learnable Context Compression for LLMs. *arXiv preprint arXiv:2503.10337*, 2025.

[4] DeepSeek-AI. DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model. *arXiv preprint arXiv:2405.04434*, 2024.

[5] Yonatan Geifman and Ran El-Yaniv. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[6] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. In *International Conference on Machine Learning (ICML)*, 2017.

[7] Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What's the Real Context Size of Your Long-Context Language Models? In *First Conference on Language Modeling (COLM)*, 2024.

[8] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM Knows What You are Looking for Before Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[9] Charles O'Neill, Alex Sandomirsky, Harry Partridge, Mudith Jayasekara, and Max Kirkby. Still: Amortized KV Cache Compaction in a Single Forward Pass. *arXiv preprint arXiv:2606.07878*, 2026.

[10] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as Operating Systems. *arXiv preprint arXiv:2310.08560*, 2023.

[11] John C. Platt. Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods. In *Advances in Large Margin Classifiers*. MIT Press, 1999.

[12] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient Streaming Language Models with Attention Sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

[13] Jingyang Yuan, Huazuo Gao, Damai Dai, Junyu Luo, Liang Zhao, Zhengyan Zhang, Zhenda Xie, Y. X. Wei, Lean Wang, Zhiping Xiao, Yuqing Wang, Chong Ruan, Ming Zhang, Wenfeng Liang, and Wangding Zeng. Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[14] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H_2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

A Evaluation-protocol record

Every certification report object embeds: (i) the item partition per axis—pointwise = needle items whose source chunk lies in the (reference) eviction window (for compactor arms: in the compacted-away region); aggregation = a disjoint same-document set; (ii) compacted-context contents (surviving KV only; no summary text); (iii) expansion disabled during certification probing; (iv) scoring by argmax over answer-letter logits under greedy decoding; (v) the KL window length. All arms share one probe partition; identical prior-correct counts across arms (21/25) verify it.

B Baseline reconciliation

Never-expand has two values in the record by design. Eviction: the earlier certification pass (0.708/0.808 raw; 0.550/0.650 conditioned) certified packed contexts with needle-anchored eviction; the canonical per-item protocol (0.453/0.551) uses hash-randomized placement (partial > 0.5 overlaps counted evicted) and raw-correct $\approx$1 denominators. Compactor: 0.271 on the CERTIFICATION protocol (Table 4: needle-anchored, needles inside the compacted region) vs 0.189 on the LABELING protocol (Table 5: hash-randomized placement, recent-raw window present). Binding rule: axis-certification claims quote the certification protocol; all operating-point, baseline, and routability claims quote the labeling protocol; the two are never mixed within one comparison.

C Decomposition, seeds, and uncertainty

Table 1 gives the three-variant decomposition (3 seeds each). Eviction headline across seeds: lost-fact incident rate 0.126 ± 0.014, expansion 0.177 ± 0.007; bootstrap over test items: $0.146\,[0.136, 0.156]$, expansion $0.168\,[0.163, 0.173]$, evicted-only AUC $0.622\,[0.608, 0.636]$. Probe arms use cluster (doc-level) bootstrap over 2,000 resamples; the MLP row is 3 seeds (SD 0.002).

D Corpus and filter survival

Four ungated sources (SEC 10-K filings; US court opinions; permissive-license source code; Project Gutenberg literature), 40,000 documents of $\geq \sim$6k tokens, $\sim$927M tokens total. Counterfactual filters retain $\approx$38% of generated items; per-domain yields 25,000/24,790/23,563/25,002 (financial/legal/code/literature). Legal-domain OCR noise is visible as the weakest per-domain gate recall.

E LODO and recalibration

Table 3 gives per-domain recalibration on $\sim$200 held-out items; discrimination transfers (pooled AUC 0.781–0.865), calibrated probabilities do not (ECE 0.05–0.13 before repair). Platt escalation restores ECE ≤ 0.035 in every rotation.

F Position-kernel collapse evidence

Cross-attention entropy 0.000 and delta-row purity 1.0 across 1,000 pilot items × 36 layers × 8 heads. Per-latent argmax source positions form the stride-≈2.998 lattice round($k \cdot (S-1)/1023$); on 5,090 re-forwarded test items the kept set is identical across heads and items (pairwise difference 0). The loss-curve-vs-routing figure (KL 0.058 by step 300, content routing 0 throughout) is stored with the capture manifest.

G Retrieval-head inventory

The four monitored heads (L25/H14 latent-mass; L26/H26, L28/H19, L35/H15 sink-mass) reach 0.70–0.72 univariate on the pilot; the 8-scalar logistic over their latent- and sink-mass reaches 0.762 on the full test split. Full inventory in `retrieval-head-inventory.json`.

H Reproducibility and compute ledger

Pinned environment (torch 2.7.1, transformers 4.55.2, safetensors 0.5.3); a 14-field compatibility fingerprint is stored with every KV artifact and validated all-or-nothing. Compute ledger: corpus ≈$500; labels ≈$110; eviction gate, multi-hop, ablations ≈$30; trained-compactor experiment ≈$715; PREPROBE ≈$62; PROBE ≈$89; FOLLOWUP ≈$9. Total ≈$1,100 marginal (corpus/gate/compactor/probes). Checkpoints, calibration records, compaction policy versions (weights hash + certified depth), evaluation grids, probe scores, and all per-item predictions are content-addressed artifacts.

I FOLLOWUP results (F1–F3)

F1 (kernel geometry). Kept-lattice extraction on 5,090 test items; $d_{\min}$ univariate AUC 0.509 $[0.499, 0.519]$, mechanics-with-kernel 0.550 $[0.537, 0.563]$ (+0.003 over $c0$), Spearman(probe score, $d_{\min}$) 0.045. Verdict: *branch (iii), smear dominates*—kernel geometry is loss-blind. **F2 (context-only control).** Refit 0.502 $[0.487, 0.518]$, zero-shot 0.508 $[0.493, 0.523]$, vs question-conditioned 0.813. Verdict: the 0.85 is a behavioral-divergence predictor, not a memory-content readout. **F3 (sink-head monitor).** 8-scalar logistic 0.762 $[0.753, 0.771]$; end-to-end lost-fact incident rate 0.149 at expansion 0.060. Ladder rung between the free self-check (0.638) and the trained stack (0.854). Verdicts, CIs, and artifacts in `FOLLOWUP-REPORT.md` and `memsub-artifacts:/probe/followup-*`.